

Power Supply

Barrel Jack 9V Power Supply to a Linear 5V Voltage Regulator with Bypass Capacitors to clean the DC Power Supply

The schematic shows a power supply circuit. On the left, a barrel jack J1 (PJ-102AH) is connected to a 1A fuse F1. The output of the fuse is connected to a 9V source through a header J2 (Conn_01x02). A test point TP1 is located between the fuse and the 9V source. The 9V source is also connected to a PWR_FLAG pin. The 9V source is connected to the input of a voltage regulator U1 (MC7805CT-BP_JEA). The regulator has three pins: Input, ground, and output. The input pin is connected to the 9V source. The ground pin is connected to Earth. The output pin is connected to a +5V output. A test point TP2 is located between the output pin and the +5V output. The +5V output is connected to a header J3 (Conn_01x02). Two bypass capacitors are shown: C1 (10uF, 50V) connected between the input and ground, and C2 (.1uF, 50V) connected between the output and ground. The text 'Edited by JEA' is present near the barrel jack.

Symbols: J2, J3 are header pins to be used for debugging, jumping, and separating sections of the schematic in case errors occur.

Light Sensor

Photoresistor in Voltage Divider Configuration, sent to a Unity Gain Buffer Op-Amp, then sent to an Inverting Amplifier Op-Amp, with a set reference voltage to give Analog Output ready for signal conditioning.

Note: Light Sensor Output is an Analog signal sent to the microcontroller to measure the sunlight the plant is receiving.

Symbols: J4, J5, J6 are jumper header pins to be used for debugging and separating sections of the schematic in case errors occur.

Microcontroller

U3
PIC18F57Q43_Curiosity_Nano_JEA

Microcontroller will be used to process signal from Light Sensor to send to the Speaker subsystem through the Motor Hub subsystem.

Symbols: C5, C6 are bypass capacitors to stabilize the DC power sent to the microcontroller

Note: Low Light Signal is a Digital High Signal sent to the Speaker subsystem to alert the user when there is low sunlight for the plant.

8-Pin Ribbon Connector

J14
Conn_02x04_Odd_Even

Low Light Signal

Earth

Note: Symbol J14 is the 8-Pin Ribbon Connector which will connect to the other subsystems as designated by the block diagram in the datasheet.

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